Import Data Transform Map

(Spreadsheet)

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PLANNING PHASE

1. Introduction:

- The Planning Phase is the initial stage of the Import Data using Transform Maps project. The main objective of this phase is to identify the requirements, define the project scope, plan the data import process, and organize the activities required to successfully import employee data into the Service Now platform.
- The project focuses on importing structured employee information from an external spreadsheet into Service Now using Import Sets and Transform Maps. The spreadsheet contains employee details such as Employee ID, Employee Name, Email, Department, and Location.
- The planning process is designed to ensure that the employee data can be imported in an organized and accurate manner. The overall process includes preparing the spreadsheet, creating the required Service Now tables, loading the data, creating the Transform Map, mapping the fields, transforming the data, validating the results, and finally generating reports and dashboards.

2. Project Objective:

- The primary objective of this project is to develop a simple and systematic method for importing employee data from an external Excel spreadsheet into ServiceNow.

The planned system aims to:

- Import structured employee data into Service Now.

- Create a dedicated Employee Test table for storing employee records.
- Use an Import Set table as a staging area for incoming data.
- Create a Transform Map between the source and target tables.
- Map spreadsheet fields to the appropriate Service Now fields.
- Insert new employee records.
- Update existing employee records.
- Prevent duplicate records using Coalesce.
- Validate imported and transformed data.
- Generate employee-related reports.
- Display important information through an Employee Analytics Dashboard.

- The project is planned as a beginner-friendly implementation that demonstrates the complete data import process in Service Now.

3. Project Scope:

- The scope of the project covers the complete employee data import and management workflow.

3.1 Data Preparation

- The first planned activity is to prepare sample employee data in a Google Spreadsheet. The spreadsheet is structured with appropriate headers and employee information. After preparation, the spreadsheet is downloaded in Sample Spreadsheet.xlsx format for importing into Service Now.

3.2 ServiceNow Table Creation:

- A custom target table named Employee Test is planned for storing the final employee records.

The planned fields include:

- Employee ID
- Employee Name
- Email
- Department
- Location

- All the fields are planned as String fields according to the project reference.

3.3 Import Set Planning:

- An Import Set table named Employee Import is planned to receive the spreadsheet data.

- The Import Set acts as an intermediate or staging area before the data is transferred to the final Employee Test table. This provides a structured path for processing the external spreadsheet information.

3.4 Transform Map Planning:

- A Transform Map named Sample Spreadsheet Import is planned between the Employee Import source table and the Employee Test target table.
- The Transform Map is responsible for defining how the source fields should be transferred to the target fields. Field mapping is planned using Auto Map Matching Fields and Mapping Assist.

4. Functional Requirements:

- The system planning includes several functional requirements that define what the project should perform.
- **Requirement 1:** Spreadsheet Data Input
  - The system should accept employee information from an Excel spreadsheet.
  - The spreadsheet should contain structured employee information with suitable headers and records.
- **Requirement 2:** Employee Table
  - The system should provide an Employee Test table for storing transformed employee records.
  - The table should contain the required employee fields such as Employee ID, Employee Name, Email, Department, and Location.
- **Requirement 3:** Data Import
  - The system should allow the spreadsheet data to be loaded into the Employee Import table.
  - The imported data should be processed before being transferred to the target table.
- **Requirement 4:** Field Mapping
  - The system should map the source fields from Employee Import to the corresponding fields in Employee Test.
- **Requirement 5:** Data Transformation
  - The system should transform the imported data and create records in the Employee Test table.
- **Requirement 6**: Duplicate Prevention

- The system should identify existing employee records using the Coalesce mechanism and avoid unnecessary duplicate records.

5. Non-Functional Requirements:

- The planning phase also considers basic qualities required for the project.

Accuracy

- The imported employee information should be transferred to the correct target fields.

Data Integrity

- The system should maintain consistent employee records during repeated imports.

Efficiency

- The Import Set and Transform Map approach should reduce the need for manually entering individual employee records.

Reliability:

- The transformation process should provide status information and transformation history so that the result can be verified.

Maintainability

- The system should use clearly defined tables, fields, mappings, reports, and dashboards so that the configuration can be managed easily.
- These requirements support the overall objective of maintaining accurate and reliable employee data in Service Now.

6. Project Activity Planning:

- The project activities are planned in a sequence so that each stage can be completed before moving to the next stage.

➢ **Phase 1** – Prepare Spreadsheet

- The sample employee data is prepared in a Google Spreadsheet and downloaded as an Excel file.

➢ **Phase 2** – Create Custom Table

- The Employee Test table is created with the required employee fields.

➢ **Phase 3** – Create Import Set Table

- The Employee Import table is created to receive the external spreadsheet data.

➢ **Phase 4** – Create Transform Map

- The Sample Spreadsheet Import Transform Map is created and configured between Employee Import and Employee Test.

➢ **Phase 5** – Transform and Validate

- The imported data is transformed and the resulting records are checked in the Employee Test table.

➢ **Phase 6** – Configure Coalesce

- Coalesce is planned to identify existing employee records and prevent duplicate entries.

➢ **Phase 7** – Import Additional Data

- Additional Excel data is planned to test whether existing employee records are updated and new employee records are inserted correctly.

➢ **Phase 8** – Create Reports

- Reports are planned for department-wise employees, location-wise employees, and employee list information.

➢ **Phase 9** – Create Dashboard

- The reports are planned to be combined into an Employee Analytics Dashboard.

7. Data Import Planning:

The data import process is planned as a controlled sequence.

Step 1: Prepare employee data in Excel.

Step 2: Load the Excel file into the Employee Import table.

Step 3: Configure the Transform Map.

Step 4: Map the source and target fields.

Step 5: Execute the Transform operation.

Step 6: Check the transformation status.

Step 7: Verify the records in the Employee Test table.

Step 8: Review Transform History.

The planning also includes testing repeated imports. When new Employee IDs are received, the system should insert new records. When existing Employee IDs are received with changed information, the system should update the existing records.

8. Duplicate Record Handling Plan:

- Duplicate handling is an important part of the project planning.
- In real-world employee data management, the same spreadsheet information may be imported multiple times. Therefore, the project plans to configure Coalesce in the Transform Map.
- The Employee ID can be used as the identifying field.

The planned logic is:

- Existing Employee ID → Update Record
- New Employee ID → Insert Record
- Same Data Imported Again → Avoid Duplicate Record

- This approach helps maintain a clean employee database. The reference project demonstrates that repeated data can result in ignored records rather than unnecessary duplicate entries.

9. Testing and Validation Plan:

- Testing is planned to confirm that the data import process works correctly.

The following test cases are included in the plan:

- Test Case 1: New Employee
- A new Employee ID is provided in the spreadsheet.
- Expected Result: A new employee record should be inserted.
- Test Case 2: Existing Employee
- An existing Employee ID is provided with updated information such as name or email.
- Expected Result: The existing employee record should be updated.
- Test Case 3: Repeated Import
- The same spreadsheet is imported again without changing the data.
- Expected Result: Duplicate records should not be created.
- Test Case 4: Transformation Status
- After transformation, the import and transformation status should be checked.
- Expected Result: The system should show the appropriate success/status information.
- Test Case 5: Final Table Verification
- The Employee Test table should be opened after transformation.
- Expected Result: The inserted and updated employee records should be available correctly.

10. Reporting and Dashboard Planning:

- The final stage of the project planning includes reporting and dashboard creation.
- Three reports are planned:
- **Report 1** – Employees by Department
- A Pie Chart is planned to display the number of employees in each department.
- Table: Employee Test
- Type: Pie Chart
- Group By: Department
- Aggregation: Count
- **Report 2** – Employees by Location
- A Bar Chart is planned to display employee distribution based on location.
- Table: Employee Test
- Type: Bar Chart
- Group By: Location
- Aggregation: Count
- **Report 3** – Employee List Report
- A List Report is planned to display detailed employee information.

The planned columns are:

- Employee ID

- Employee Name
- Email
- Department
- Location

- These reports are then planned to be added to the Employee Analytics Dashboard so that employee information can be viewed from a centralized location.

11. Expected Outcome:

- At the end of the planned project, the system is expected to provide a complete employee data import workflow.

The expected workflow is:



- The planned implementation should allow employee information to be imported, transformed, updated, validated, and presented through reports and dashboards.

- The reference project also extends the solution with reports for total employee records, newly added employees, updated employee records, department-wise distribution, and import success/error information.

12. Planning Phase Summary:

- The Planning Phase establishes the complete roadmap for the Import Data using Transform Maps project. It identifies the source spreadsheet, target Employee Test table, Employee Import staging table, Transform Map, field mappings, Coalesce configuration, validation process, reports, and dashboard.
- Proper planning ensures that the project follows a clear sequence from data preparation to final visualization. It also prepares the system to handle both new and existing employee records while reducing duplicate data.
- Therefore, the planned solution provides a structured approach for importing and managing employee information in Service Now using Import Sets and Transform Maps.